

Big data Analytics:-

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Q Describe the various major components in hadoop ecosystem with examples:-

Ans: The Hadoop ecosystem is a framework for processing and managing large datasets across distributed clusters. It consists of several components, each serving specific purpose. The major components of Hadoop ecosystem are:-

1. Hadoop distributed file system (HDFS):-

Function:- A distributed storage system that splits large files into smaller blocks and distributes them across multiple nodes in cluster.

Eg: If a 10 GB file is stored in HDFS with a block size of 128 MB, it gets divided into 8 blocks & distributed across nodes.

2. Yet Another Resource Negotiator (YARN)

Function:- Manages resources & schedules tasks across the cluster.

Eg: If multiple app (mapreduce, spark) runs on same cluster, YARN ensure sufficient resource allocation.

3. MapReduce:- A programming model for processing large-scale data in parallel.

Example: word count in a large dataset.

Map phase:- Split text into words & assign 1 to each word.

Reduce phase:- Aggregate word counts.

4. Apache Hive:-

Function:- A data warehouse system that allows querying structured data using SQL like language (HiveQL).

Core Principles of Hadoop:-

① Scale out rather than scale up.

- It's harder & more expensive to scale up.
 - Add additional ~~its~~ resources to an existing CPU.
 - New units must be purchased if required resources cannot be added.
- Also known as Scale vertically.

→ Scale out

- Add more nodes to an existing distributed application.
- Software layer is designed for nodes addition or removal.
- Very easy to scale down as well.

② Bring code to data instead of data to code:-

- Hadoop locate collocate processors & storage.
- Code is moved to data.
- Processes execute code & access underlying local storage.

③ Hadoop is designed to cope with node failures

→ Data replication

- If a node fails master will detect the failure & assign the task to new node.
- If a failed node restarts, it automatically joined back to the app & assigned new task.
- If a node is running slow, the master can redundantly execute another instance of same task.

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Ex → Running SQL queries to analyze logs stored in HDFS

5. Apache HBase :-

Function :- A NoSQL database that provides real-time read/write access to large datasets

Example :- Storing user clickstream data for real-time analytics

Other components are :- Apache Pig, Apache Sqoop, Apache Flume, Apache Oozie, Apache Zookeeper, Apache Mahout

Q2 List the core principles of Hadoop ? How the features are handled by Hadoop?

→ Core Principles :-

1) Distributed Storage (HDFS) :-

→ data is distributed split into blocks and distributed across multiple nodes

→ Provides fault tolerance through data replication

2) Parallel processing (MapReduce) :-

→ Data processed in parallel across nodes

→ uses divide and conquer approach for fast computation

3) Scalability :-

→ Can handle petabytes of data by simply adding more nodes

4) Fault Tolerance :-

→ data is replicated across multiple nodes

→ If a node fails another copy of data is used automatically

How Hadoop handles failures:-

1) Data duplication in HDFS:-

- HDFS stores multiple copies (3) ^{of} each data on different nodes.
- If a node crashes, it automatically retrieves data from other nodes.

2) NameNode failure mechanism:-

- Primary NameNode keeps metadata, Secondary NameNode helps in
- on high availability mode, two NameNodes run in active ^{standby} standby state.

3) TaskTracker & Job Tracker:-

- If a TaskTracker node fails, the JobTracker reassigns the task to another node.

4) Resource Manager & Node Manager.

- Resource Manager monitors & reschedules tasks in case of failure.
- If NodeManager, running tasks are reassigned to another node.

Q Two main Abstractions for parallel programming?

1) Resilient Distributed Dataset (RDD):-

- An immutable collection of distributed objects across a cluster.

Features:-

Lazy Evaluation:- Operations are not executed immediately but when action is triggered.

Fault-tolerant:- Uses lineage to compute lost partitions in case of failure.

2) DataFrame:-

- A distributed collection of structured data.

Features

→ **Schema Based:** has named columns like SQL.

→ **Optimized execution:-** Uses Catalyst optimizer & Tungsten engine for efficiency.

→ allow Spark SQL integration
 Example - Analyzing sales data stored in database

→ Two common RDD transformations

1) Map(): Transformation

→ applies a function to each element & returns a new RDD

$rdd = [1, 2, 3, 4, 5]$

$square_rdd = rdd.map(\lambda x: x^2)$

2) Filter(): Transformation:-

→ Selects element that satisfy a condition

$rdd = [10, 15, 20, 25, 50]$

$filtered_rdd = rdd.filter(\lambda x: x > 20)$

Q Comparison of different types of NoSQL databases -

TYPE	Description	Data-model	Best used for	Example
→ Key-value Store	Stores as key-value pair	hash table (key-value)	Caching, session management	Redis, Amazon DynamoDB
→ Doc. Store	Stores data as JSON like doc	JSON/BSON doc	Content management user profiles	MongoDB, CouchDB
→ Column-Family stores	Stores data in col. instead of rows	column oriented	Big data analytics	Apache Cassandra
→ Graph databases	Stores relationships b/w data as nodes	Graph	Social network fraud detection	Neo4j, Amazon Neptune

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Q MapReduce Programming model:-

→ MapReduce is a distributed data processing model using Apache Hadoop. to process large datasets in parallel. It follows a divide and conquer approach with two key phases:-

① Map Phase (data processing)

- The input dataset split into smaller key-value pair
- The mapper function process each key value & emits immediate key value

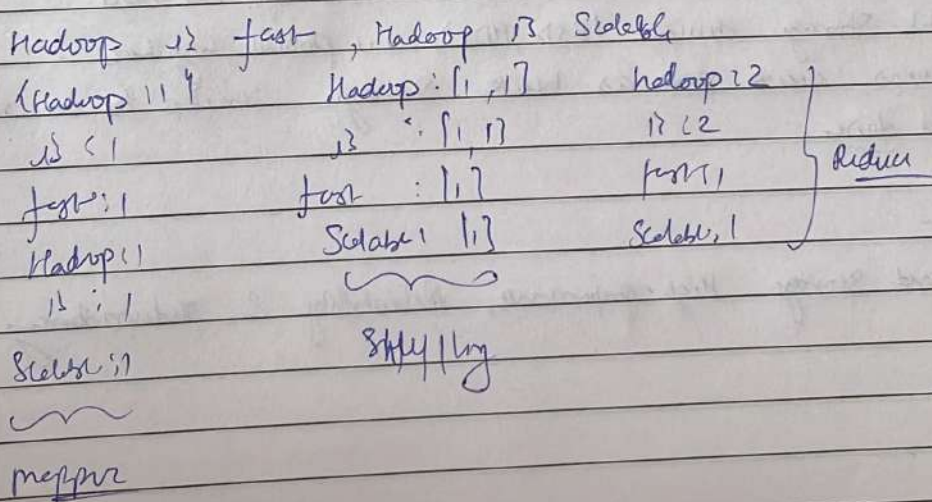
② Shuffle & Sort (Data transfer)

- The intermediate key-value pairs are sorted & grouped by key.
- This steps helps prepare data for next phase

③ Reduce phase (Aggregation)

- The reducer function processes grouped key-value pairs
- It aggregates the data & produces the final output

Q word count:-



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Q NAS And SAN

→ Network attached storage:

- A file-based storage system connected to a network
- Allows multiple users and devices to access shared files over ethernet.

→ Functions like dedicated file server.

How it works -

- Uses protocol like NFS (Linux), SMB/CIFS (Windows)
- Data stored in centralized storage unit, accessible to multiple client.

Features -

→ File Based Storage, Easy Setup, Scalability, Cost-effective.

→ Storage Area Network - (SAN)

→ A block-level storage solution for high-performance applications.

→ Uses a dedicated storage network instead of ethernet.

→ Appears as a local drive to servers rather than shared network storage.

How it works?

→ Uses protocol like Fibre channel to transfer the data.

→ SAN Storage devices (SSD, HDD) are pooled together.

→ Servers access SAN as block storage similar to an internal hard drive.

Features -

Block-Based Storage, High-performance, Reliability & redundancy.